

Yusi Wei

✉ yw825@drexel.edu

✉ weiyusio@gmail.com

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Education

- 2021-2026 **Ph.D., Drexel University** Operations & Business Analytics (3.62/4.0 GPA)
Advised by Hande Benson and Muge Capan
Graduate Minor in Computer Science (3.80/4.0 GPA)
- 2019-2021 **M.S., University of South Florida** Business Analytics & Information Systems (3.97/4.0 GPA)
- 2015-2019 **B.S., Jiangnan University** Information Systems & Information Management (3.56/4.0 GPA)

Research Overview

- Identify privacy-vulnerable patients in healthcare data, advancing responsible AI in clinical settings
- Develop optimization models to balance privacy protection and predictive accuracy in ML applications
- Aim to study patient trust in AI-driven healthcare toward building trustworthy AI systems

Papers

An Analytical Approach to Privacy and Performance Trade-Offs in Healthcare Data Sharing

Under second-round review at Healthcare Analytics

With Hande Benson, Muge Capan

- Identified that underrepresented subgroups — older adults, frequently hospitalized patients, and minorities — face up to $5\times$ higher risk of the most common types of privacy attacks in the healthcare dataset
- Demonstrated that excluding these subgroups reduces ML precision and recall by up to 40%, establishing a fundamental privacy-utility tension in healthcare data sharing
- Evaluated state-of-the-art anonymization techniques, achieving protection for up to 80% of vulnerable individuals while maintaining $\sim 90\%$ ML performance

Multi-Objective Optimization-Based Anonymization of Structured Data for Machine Learning

Under third-round review (Minor revision) at Data Mining and Knowledge Discovery

With Hande Benson, Joseph K. Agor, Muge Capan

- Building on identified vulnerable populations, proposed a multi-objective optimization model that jointly minimizes information loss and maximizes protection against privacy attacks, using information loss as a surrogate measure of ML performance
- Achieved protection for over 90% of vulnerable individuals while maintaining $\sim 90\%$ predictive performance of the healthcare dataset
- Demonstrated generalizability by validating on financial and census datasets, confirming consistent privacy-utility improvements beyond the healthcare domain

A Closed-Loop Framework for Data Anonymization in Machine Learning Applications

Under review at Operations Research

With Hande Benson, Muge Capan

- Motivated by the surrogate nature of information loss in the prior model, proposed a closed-loop optimization framework that directly incorporates ML performance as feedback to iteratively refine privacy-preserving strategies
- Achieved zero privacy attack risk against the most common attack types for individuals in under-represented groups, while protecting over 90% of individuals against more advanced attack types
- Demonstrated that directly optimizing ML performance, rather than using a surrogate, improves predictive performance by $\sim 10\%$, validating the closed-loop approach over static optimization

Teaching

Instructor

- 2026 **Introduction to Data Mining for Business**, STAT 331 (Undergraduate course)
Statistical and ML models for business decision-making and data-driven insights
- 2025 **Programming for Data Analytics in Python**, BSAN 360 (Undergraduate course)
Teaching evaluations: 3.57/4.0, 3.63/4.0
Python programming for data analysis, visualization, and AI-assisted coding
- PhD Statistics Bootcamp**
Statistics overview for incoming doctoral students emphasizing research applications
- 2024 **Programming for Data Analytics in R**, BSAN 360 (Undergraduate course)
Teaching evaluation: 3.07/4.0
R programming for data analysis and visualization
- 2022, 2023 **Management Information Systems Recitation**, MIS 200 (Undergraduate course)
Data analysis in Excel and database design in Microsoft Access

Teaching Assistant

- 2021–2026 **Capstone:** Business Analytics Capstone Project (BSAN 710)
MBA: Business Analytics for Managers (BSAN 610); Database Analysis and Design (MIS 632); Statistics for Business Analytics (STAT 610)
MS: Supply Chain Management II (POM 615); Managerial Decision Models and Simulation (OPR 601); Data Mining for Managers (STAT 632)
Undergraduate: Linear Models for Decision Making (OPR 320); Operations Management (OPM 200); Business Statistics II (STAT 202); Business Analytics and Data Visualization (BSAN 160)

Service

- 2025 **INFORMS, Job Market Showcase**, Session Chair
Journal of Optimization Theory and Applications, Reviewer
Workshop on Information Technologies and Systems, Reviewer; Discussant
Capstone Class, Faculty Panel
- 2024 **INFORMS, OPT: Computational Optimization and Software**, Session Host
- 2024, 2023, 2022 **Drexel University, LeBow Tutoring & Learning Center**, Tutor

Presentations

A Closed-Loop Framework for Data Anonymization in Machine Learning Applications
2025 Workshop on Information Technologies and Systems

Trade-Offs Between Privacy and Utility in Machine Learning

2025 Finalist, INFORMS MIF competition

2025 INFORMS Annual Meeting

2025 Philadelphia Operations and Technology Day conference

2025 FutureBAProf Workshop at University of Iowa

Multi-Objective Optimization-Based Anonymization of Structured Data for Machine Learning

2024 Princeton Workshop on Optimization, Learning, and Control

2024 INFORMS Annual Meeting

References

Hande Benson

Professor of Decision Sciences and MIS
Lebow College of Business, Drexel University
3220 Market Street, Gerri C. Lebow Hall 743
Philadelphia, PA, 19104
Phone: 215-895-6999
Email: hvb22@drexel.edu

Muge Capan

Assistant Professor of Mechanical & Industrial Engineering Department
University of Massachusetts-Amherst
213D Engineering Laboratory
160 Governors Drive
Amherst, MA, 01003
Phone: 413-545-5735
Email: mcapan@umass.edu

Qizhi Dai

Associate Professor of Decision Sciences and MIS
Lebow College of Business, Drexel University
3220 Market Street, Gerri C. Lebow Hall 620
Philadelphia, PA, 19104
Phone: 215-895-2121
Email: qd24@drexel.edu

Arjan Raven

Associate Clinical Professor of Decision Sciences and MIS
Lebow College of Business, Drexel University
3220 Market Street, Gerri C. Lebow Hall 738
Philadelphia, PA, 19104
Phone: 215-895-2134
Email: ar3783@drexel.edu